KAIZEN MATHS

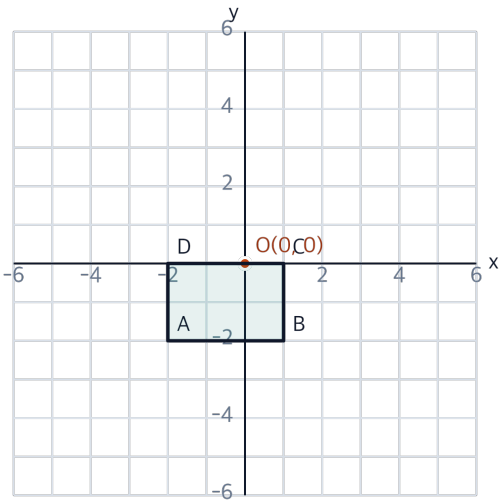
Lesson starter

Transformations

Rotate & Enlarge

Name _____

SA1



Rotate shape ABCD 90 degrees clockwise about the origin. Give the coordinates of the image.

GEO-C8E6-L2 · SA

Show your working

KAIZEN MATHS

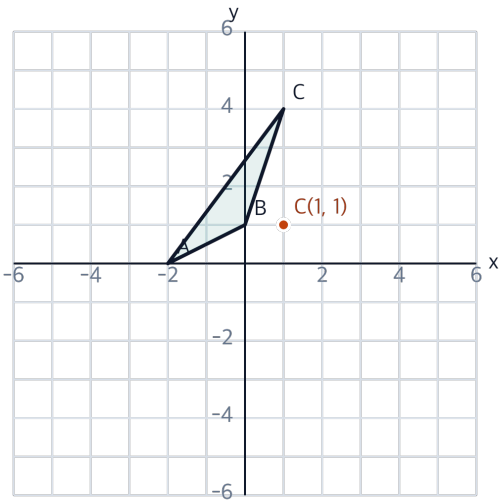
Lesson starter

Transformations

Rotate & Enlarge

Name _____

SB1



Rotate shape ABC 90 degrees anticlockwise about centre (1, 1). Give the coordinates of the image.

GEO-C8E6-L2 · SB

Show your working

KAIZEN MATHS

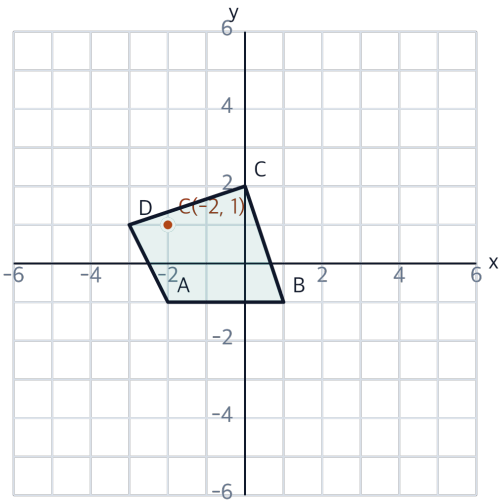
Lesson starter

Transformations

Rotate & Enlarge

Name _____

SC1



Enlarge shape ABCD by scale factor 2 from centre (-2, 1). Give the coordinates of the image.

GEO-C8E6-L2 · SC

Show your working

KAIZEN MATHS

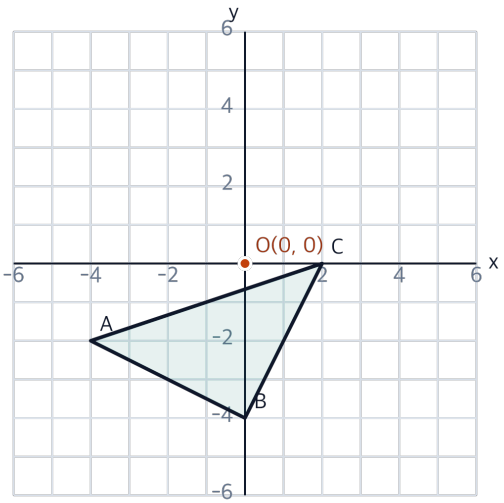
Lesson starter

Transformations

Rotate & Enlarge

Name _____

SD1



Enlarge shape ABC by scale factor $\frac{1}{2}$ from centre (0, 0). Give the coordinates of the image.

GEO-C8E6-L2 · SD

Show your working

KAIZEN MATHS

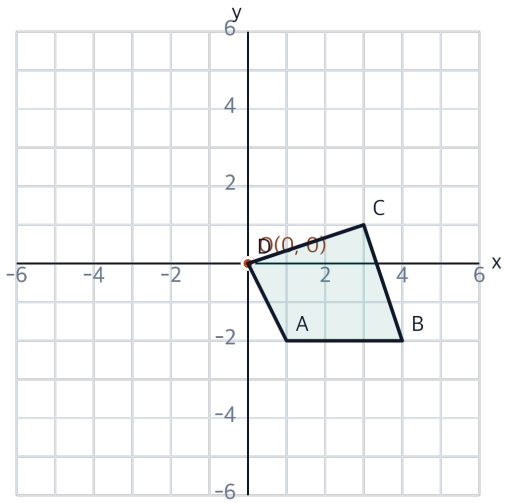
Exit ticket

Transformations

Rotate & Enlarge

Name _____

EA1



Rotate shape ABCD 90 degrees anticlockwise about the origin. Give the coordinates of the image.

GEO-C8E6-L2 · EA

Show your working

KAIZEN MATHS

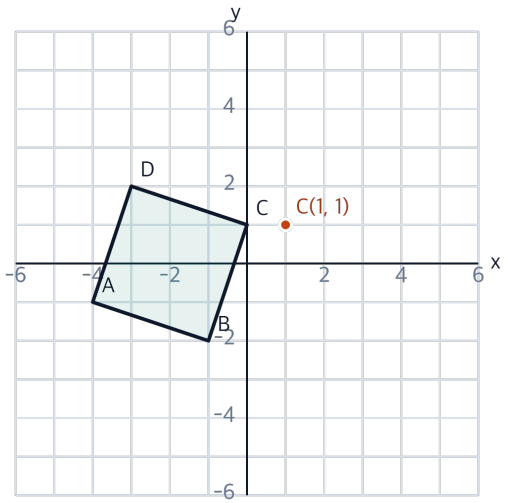
Exit ticket

Transformations

Rotate & Enlarge

Name _____

EB1



Rotate shape ABCD 90 degrees clockwise about centre (1, 1). Give the coordinates of the image.

GEO-C8E6-L2 · EB

Show your working

KAIZEN MATHS

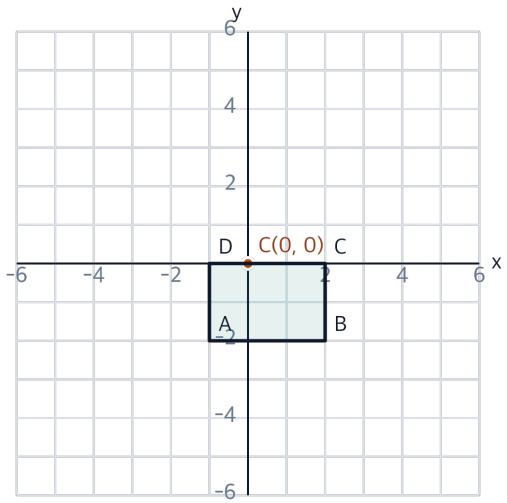
Exit ticket

Transformations

Rotate & Enlarge

Name _____

EC1



Enlarge shape ABCD by scale factor 3 from centre (0, 0). Give the coordinates of the image.

GEO-C8E6-L2 · EC

Show your working

KAIZEN MATHS

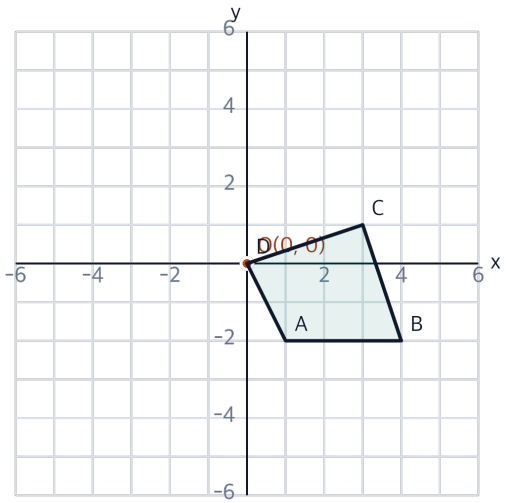
Exit ticket

Transformations

Rotate & Enlarge

Name _____

EA1



Rotate shape ABCD 90 degrees anticlockwise about the origin. Give the coordinates of the image.

GEO-C8E6-L2 · EA

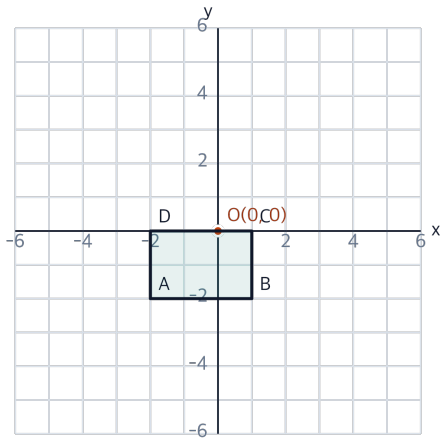
Show your working

Teacher answers & working

Transformations · Rotate & Enlarge

GEO-C8E6-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Rotate shape ABCD 90 degrees clockwise about the origin. Give the coordinates of the image.

Answer:

A'(-2, 2), B'(-2, -1), C'(0, -1), D'(0, 2)

Working:

The centre of rotation is the origin (0, 0).

For a 90 degrees clockwise rotation about the origin, use the rule (x, y) → (y, -x).

Vertex	Original point	Rule	Image point
A	(-2, -2)	(x, y) → (y, -x)	A' (-2, 2)
B	(1, -2)	(x, y) → (y, -x)	B' (-2, -1)
C	(1, 0)	(x, y) → (y, -x)	C' (0, -1)
D	(-2, 0)	(x, y) → (y, -x)	D' (0, 2)

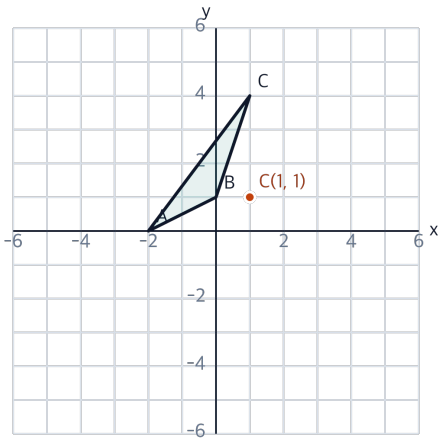
Final image coordinates: A'(-2, 2), B'(-2, -1), C'(0, -1), D'(0, 2).

Teacher answers & working

Transformations · Rotate & Enlarge

GEO-C8E6-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Rotate shape ABC 90 degrees anticlockwise about centre (1, 1). Give the coordinates of the image.

Answer:

A'(2, -2), B'(1, 0), C'(-2, 1)

Working:

Use centre (1, 1). Work with the displacement from the centre to each vertex.

Rotate each displacement 90 degrees anticlockwise, then add it back to the centre.

Vertex	Original	Vector from centre	Rotated vector	Image
A	(-2, 0)	(-3, -1)	(1, -3)	A' (2, -2)
B	(0, 1)	(-1, 0)	(0, -1)	B' (1, 0)
C	(1, 4)	(0, 3)	(-3, 0)	C' (-2, 1)

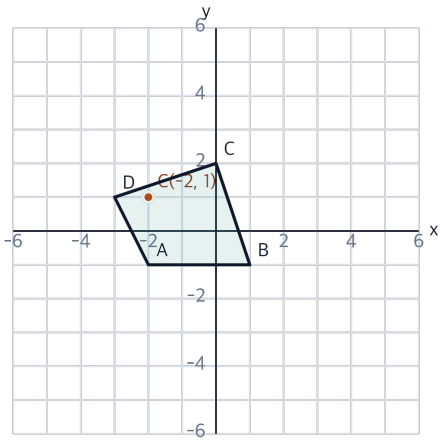
Final image coordinates: A'(2, -2), B'(1, 0), C'(-2, 1).

Teacher answers & working

Transformations · Rotate & Enlarge

GEO-C8E6-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Enlarge shape ABCD by scale factor 2 from centre (-2, 1). Give the coordinates of the image.

Answer:

A'(-2, -3), B'(4, -3), C'(2, 3), D'(-4, 1)

Working:

Start from centre (-2, 1).

For each vertex, find the vector from the centre and multiply it by scale factor 2.

Vertex	Original	Vector from centre	Scaled vector	Image
A	(-2, -1)	(0, -2)	2(0, -2) = (0, -4)	A' (-2, -3)
B	(1, -1)	(3, -2)	2(3, -2) = (6, -4)	B' (4, -3)
C	(0, 2)	(2, 1)	2(2, 1) = (4, 2)	C' (2, 3)
D	(-3, 1)	(-1, 0)	2(-1, 0) = (-2, 0)	D' (-4, 1)

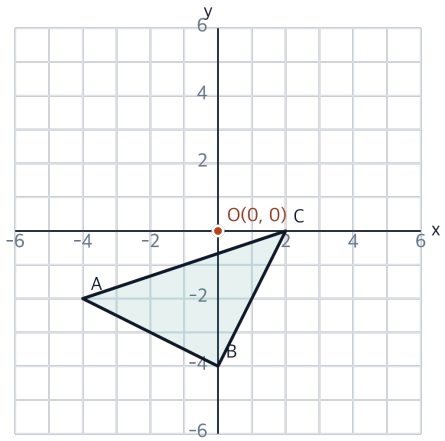
Final image coordinates: A'(-2, -3), B'(4, -3), C'(2, 3), D'(-4, 1).

Teacher answers & working

Transformations · Rotate & Enlarge

GEO-C8E6-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



Enlarge shape ABC by scale factor $\frac{1}{2}$ from centre (0, 0). Give the coordinates of the image.

Answer:

A'(-2, -1), B'(0, -2), C'(1, 0)

Working:

The centre is the origin, so each position vector is multiplied by $\frac{1}{2}$.

The shape becomes smaller because the scale factor is between 0 and 1.

Vertex	Original	Vector from centre	Scaled vector	Image
A	(-4, -2)	(-4, -2)	1/2(-4, -2) = (-2, -1)	A' (-2, -1)
B	(0, -4)	(0, -4)	1/2(0, -4) = (0, -2)	B' (0, -2)
C	(2, 0)	(2, 0)	1/2(2, 0) = (1, 0)	C' (1, 0)

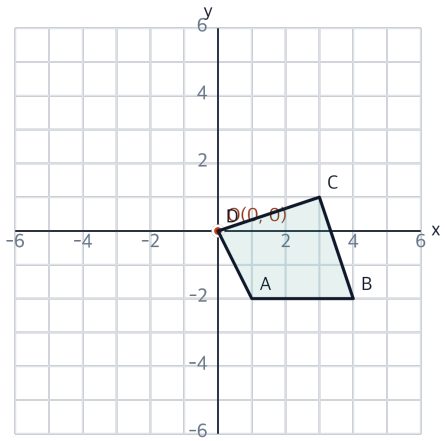
Final image coordinates: A'(-2, -1), B'(0, -2), C'(1, 0).

Teacher answers & working

Transformations · Rotate & Enlarge

GEO-C8E6-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Rotate shape ABCD 90 degrees anticlockwise about the origin. Give the coordinates of the image.

Answer:

A'(2, 1), B'(2, 4), C'(-1, 3), D'(0, 0)

Working:

The centre of rotation is the origin (0, 0).

For a 90 degrees anticlockwise rotation about the origin, use the rule (x, y) → (-y, x).

Vertex	Original point	Rule	Image point
A	(1, -2)	(x, y) → (-y, x)	A' (2, 1)
B	(4, -2)	(x, y) → (-y, x)	B' (2, 4)
C	(3, 1)	(x, y) → (-y, x)	C' (-1, 3)
D	(0, 0)	(x, y) → (-y, x)	D' (0, 0)

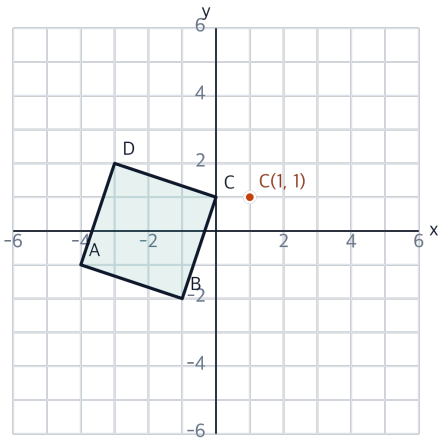
Final image coordinates: A'(2, 1), B'(2, 4), C'(-1, 3), D'(0, 0).

Teacher answers & working

Transformations · Rotate & Enlarge

GEO-C8E6-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Rotate shape ABCD 90 degrees clockwise about centre (1, 1). Give the coordinates of the image.

Answer:

A'(-1, 6), B'(-2, 3), C'(1, 2), D'(2, 5)

Working:

Use centre (1, 1). Work with the displacement from the centre to each vertex.

Rotate each displacement 90 degrees clockwise, then add it back to the centre.

Vertex	Original	Vector from centre	Rotated vector	Image
A	(-4, -1)	(-5, -2)	(-2, 5)	A' (-1, 6)
B	(-1, -2)	(-2, -3)	(-3, 2)	B' (-2, 3)
C	(0, 1)	(-1, 0)	(0, 1)	C' (1, 2)
D	(-3, 2)	(-4, 1)	(1, 4)	D' (2, 5)

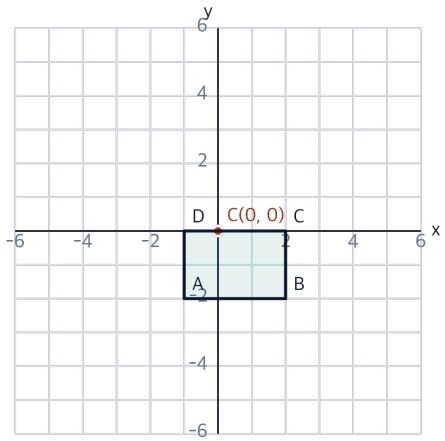
Final image coordinates: A'(-1, 6), B'(-2, 3), C'(1, 2), D'(2, 5).

Teacher answers & working

Transformations · Rotate & Enlarge

GEO-C8E6-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Enlarge shape ABCD by scale factor 3 from centre (0, 0). Give the coordinates of the image.

Answer:

A'(-3, -6), B'(6, -6), C'(6, 0), D'(-3, 0)

Working:

Start from centre (0, 0).

For each vertex, find the vector from the centre and multiply it by scale factor 3.

Vertex	Original	Vector from centre	Scaled vector	Image
A	(-1, -2)	(-1, -2)	3(-1, -2) = (-3, -6)	A' (-3, -6)
B	(2, -2)	(2, -2)	3(2, -2) = (6, -6)	B' (6, -6)
C	(2, 0)	(2, 0)	3(2, 0) = (6, 0)	C' (6, 0)
D	(-1, 0)	(-1, 0)	3(-1, 0) = (-3, 0)	D' (-3, 0)

Final image coordinates: A'(-3, -6), B'(6, -6), C'(6, 0), D'(-3, 0).